

Handover: the 2026-08-27 arc — review, lane entry, fitting preparation, assessment

Written 2026-08-27 for a successor session; extended the same evening after a live discussion with Jonas. Read `handover.md` and `handover_2026-08-26.md` first; this file covers only this arc. Sections 1–6 cover the day's work (review, lane entry, fitting preparation); sections 7–9 cover the evening (the honest assessment, the discussion rounds that amended the plan, and — read this if you are the next session — the model-handover protocol, since the next session is expected to run on a less expensive model executing `docs/czb_work_orders.md`). Jonas intends to hand the entire chat transcript to the next session alongside this file; where transcript and repository disagree, the repository wins.

1 The review: what was checked and what it found

The previous arc's own §9 flagged two things to verify; both were run, and the review went beyond them. Findings, in decreasing order of consequence:

1.1 The B-versus-C "statistically solid" claim was overstated, and is now corrected. The paired-difference analysis behind results doc §4.3b-ii ($\theta_C - \theta_B = 0.061$, 95% HDI [0.037, 0.107], "same sign on every resample") had no committed script. Reproducing it revealed it had resampled the reference while holding **both synthetic sides fixed**, omitting the synthetic-side sampling variance — which is independent between conditions and does not cancel in the difference. The committed reproduction (`replication/causation/paired_difference.py`, output `output/summary_paired_difference_fullp_abn.md`) runs all three schemes: `ref-only` reproduces the old numbers; under the project's own population convention the result is mean 0.055, 95% HDI [−0.006, 0.110], $P(\theta_C > \theta_B) \approx 0.97$. The ordering is strongly supported — consistent in point estimates at every bin count, corroborated by the braking aggregate (B within 0.3%, C over-braking 35%), ~0.97 posterior probability — but not "every resample". Corrected in results doc §4.3b-ii and §8, handbook ch. 08, README, and a bracketed correction in `handover_2026-08-26.md` §3.6. The irony that the arc devoted to uncertainty accounting produced one more incomplete uncertainty accounting is noted here so a successor keeps checking.

1.2 The stored bin-sensitivity summary and the results doc quoted intervals from different resampling conventions, without saying so. The stored sweep predated the population default; the doc's $N = 20$ remark quoted a third, intermediate run whose HDI matched neither. The sweep is regenerated under the population convention (its header now states the convention), and the θ -bias remark in §4.4 is corrected: under cases resampling B's $N = 20$ HDI [0.347, 0.773] excludes the point estimate 0.275; under the population convention it is [0.263, 0.389] and does not.

1.3 The §4.3 hand-copied intervals check out. Every θ , Θ , HDI and aggregate cell in §4.3/4.3b/4.3c matches `summary_fullp_abn.md` exactly.

1.4 Code defects found and fixed (none affected published numbers, verified): `equivalence_test` truncated weights to the filtered length instead of applying the finite mask — misaligning every weight-value pair after a mid-array NaN; all published inputs were checked NaN-free, so this never fired; fixed with a regression test. The resample error message omitted "population". `cutin_norm_weight`'s in-lane and adjacent bands tiled the whole corridor, leaving the straddling category empty by construction — the time-dependent norm could never engage; fixed (still neutral by default). `bin_sensitivity.py`'s `--abnormal` flag could never be turned off.

1.5 Role assignment misbehaves outside cut-ins, as §9 suspected: on the overtake traces it picks the cyclist as ego; on the truck-lateral traces (no vehicle moves laterally) the choice is arbitrary and picks a car 30 m behind. Cut-in traces (car and truck) assign correctly; LTAP finds the right target but the cut-in x/y conventions are meaningless in crossing geometry. Documented as a prerequisite of the transfer test (`docs/czb_fitting_plan.md` §5).

1.6 Documentation staleness fixed: duplicate section number 4.4 in the results doc (bin sweep and counterfactual rule; renumbered 4.5/4.6 with cross-references); the `handover.md` §2 row that said " $\Theta = 0.05$ derived from an aggregate-error bound" (the adopted value is 0.089; 0.05 is what a 5% tolerance would give); stale test counts.

2 The continuous lane-entry form (settled) and the staging fix

The full argument is `docs/lane_entry_note.md`; handbook ch. 04's open-problem section now records the resolution (`{{R4}}`, dark purple). In one paragraph: the binary lane gates are replaced by the **lateral overlap fraction predicted**

at the moment of longitudinal closure (the lateral offset projected forward at its closing rate over the longitudinal TTC, clamped at centering — Jonas's τ_{lat} -versus-TTC comparison expressed as an overlap), and the saturated safety magnitude by the **counterfactual residual impact speed** $\sqrt{\max(0, v_{react}^2 - 2 a_{max} d_{avail})}$, which is zero exactly at the model's own boundary $a_{req} = -a_{max}$ and turns the cliff into a ramp. Both changes are parameter-free; both reduce exactly to the released forms in every geometry the released scenarios sustain; both are flag-gated in `src/aidriver/preferences.py` (`lane_entry_continuous`, `counterfactual_residual_severity`, defaults off) so every published number is untouched. The note argues the reading on which this is the closed-form of the expectation the closed loop already computes by rollout, not a new preference. The desired speed is staged from the clip in `cutin_params` — the released scenarios stage `v_ego_des` the same way.

Validation (`replication/czb/cutin_field_check.py`, `figures/cutin_field_check.png`): pre-onset deficit exactly zero in every clip; deficit ordered by criticality at every truncation point C2–C6 for cars and trucks; ramps, not steps, within the response window. 19 new property tests (`tests/test_cutin.py`) including the released-limit identities and a continuity test that caught a genuinely discontinuous first design (recorded in the note as a discarded intermediate).

Two empirical findings worth carrying:

- **The mild truck cut-ins (TTC6–8) have zero dread-field deficit through most of the response window** — at those gaps the counterfactual crash is avoidable at $a_{max} = 8$, so the field is silent where humans plausibly still press. The comfort–dread distinction (allowed deceleration below capability) is therefore a necessity for the fitting, not a refinement.
- **Onset detection was 1–2 frames late** (2%-of-span threshold), which put genuinely moving frames on the pre-onset side of the C1 anchor; fixed to an absolute 0.03 m threshold. Found via the C1 covariate check, which is the kind of thing only the data pipeline surfaces.

3 Fitting preparation (done) and the methods answer

`docs/czb_fitting_plan.md` is the document; `src/comfortzone/czb_data.py` the pipeline; `replication/czb/pilot_surface_fit.py` the committed stage-0 fit.

The answer to "do we have optimal methods with so many parameters": the fear is misplaced, by design. Everything upstream of the boundary is frozen (released parameters; parameter-free structural changes; staging from the clip), the stimulus field is precomputed once per clip, and all 3 096 Random cut-in trials share 18 covariate values. What remains is a generalized linear mixed model — ~5–8 population parameters plus shrunk per-driver random effects — for which MAP + Laplace in CPU-torch is entirely adequate, with a synthetic-recovery harness as the first step. No simulator in the likelihood, no CMA-ES, no ABC.

Stage-0 feasibility, already run: a two-parameter probit on the running-max deficit reproduces the study's 18-cell response surface with **correlation 0.90, RMSE 0.125** — and its residual (too flat in time within a criticality level) is precisely the accumulator's signature, motivating stage 2 before it is fitted. The Button pipeline works too: 2 396 trials, press times on the time-since-onset axis, medians ordered by criticality (TTC4 0.36 s → TTC8 1.8 s), only 0.5% never press.

4 What changed on disk

New: `docs/lane_entry_note.md`, `docs/czb_fitting_plan.md`, `handover_2026-08-27.md` (each with Word/PDF), `src/comfortzone/czb_data.py`, `tests/test_cutin.py`, `replication/czb/{cutin_field_check,pilot_surface_fit}.py`, `replication/causation/paired_difference.py` (+ its summary and log in out/), `figures/cutin_field_check.png`. Modified: `src/aidriver/preferences.py` (`flags` + `lane_entry_weight` + `residual_delta_v`), `src/comfortzone/cutin.py` (staging, obs fields, onset fix, norm-weight fix), `src/equivalence/test.py` (weight mask, error message), `replication/causation/bin_sensitivity.py`, `tests/test_causation.py` (regression test), `docs/crash_causation_results.md` (§4.3b-ii rewritten, §4.4 corrected, renumbering), `handbook chs. 04/08` + `build_handbook.py` ({{R4}} dark purple), `README.md`, `handover.md`, `handover_2026-08-26.md` (correction note), regenerated `out/summary_bin_sensitivity_fullp_abn.md`. Suite: 123 tests, all passing.

5 Where I would pick up

- 1 **Synthetic-recovery harness, then the stage-1 hierarchical fit** on the Random cut-in surface (fitting plan §2–3). Jonas's four questions in plan §6 gate stage 1 but not the harness.
- 2 **Add the truck cut-in cells** and run the deficit-versus-allowed-deceleration check (plan §4) — the place the data may overrule the scalar-level framing.
- 3 **The accumulator layer**, fitted on Random, validated on Button press-time densities with one paradigm shift.
- 4 **Transfer-scenario field constructions** — LTAP crossing geometry, role assignment by instructed-vehicle for overtake/truck-lateral (finding 1.5).
- 5 Standing items unchanged: Jonas's handbook review (now R1–R4), the Jian Wu email, the $a_{f,min}$ execution model, the epistemic gaze half.

6 Caveats on the day sections

Written by the session that did the work. The stage-0 fit is population-level and deliberately naive; do not quote its c or σ as boundary estimates. The C1 covariate cells are small but nonzero for TTC6/8 (the anticipation weight activates on the first frame of motion); the design's bias term absorbs this, but anyone comparing C1 predictions to the "pure baseline" reading should know it is not exactly pure. The Button-side covariates at the press assume the clip's field applies up to the press frame; the 23 overshoot trials are dropped by the loader, per the dataset's own guidance.

7 The evening: assessment and the discussion that amended the plan

Jonas asked for an honest assessment of active inference for CZB quantification, then pushed on it in two rounds. Everything below is committed; the documents carry the full arguments.

The assessment (docs/active_inference_for_czb_assessment.md): the method uses a thin slice of the framework (preference field + accumulator; planning and epistemic value off); the distinctive value lives in the untested transfer claim and the dual role (one scalar defines the boundary *and* predicts response timing); seven risks with severities; for ADAS triggering the percentile choice dominates and comfort is an acceptance criterion. Verdict: pursue, transfer test as the gate, describe more modestly than the "active inference" label suggests.

Numbers produced during the discussion (all committed under replication/czb/):

- *Baselines and cross-validation* (pilot_surface_fit.py): the field beats every single kinematic scalar at matched parameter count but loses within-scenario to richer regressions, and the result is NOT overfitting — held-out (leave-one-criticality-out) the field gives corr 0.864 against **Jonas's 2D state rule** (probit on $1/TTC + \text{lateral offset}$) at **0.921**, now the named comparator. The 2D rule's second observable is scenario-specific, so the transfer test is a three-way race: one-scalar field vs per-scenario 2D rules vs the elliptical joint.
- *Perceived safety* (ps_vs_field.py, figures/ps_vs_field.png): the PS column is perceived **unsafety** (dictionary gotcha 1). Against the field: cell-level Spearman $\rho = 0.93$, trial-level 0.58, within-criticality 0.39–0.64. The ratings keep rising where the static field compresses — the accumulator's signature a third time.
- *Epistemic value, from the authors' own ablations* (epistemic_ablation_check.py): the deposit contains $\alpha = 0$ runs. Rear-end, 28 matched configurations: collision difference exactly 0.000; intersection within 0.005; oncoming 5.7 points *fewer* collisions with the term on (3 configurations). Inert longitudinally — our $\alpha = 0$ reproduces the authors' own ablation — possibly useful laterally.
- *Boundary sensitivity*: relative speed is in the closed form (comfort boundary at 20 m/s: 1.86 s steady $\rightarrow$ 2.59 s closing at 5 m/s); the 1.86 stacks three worst-case conventions ($a_{OV,min} = -4$ alone gives 1.03 s).
- *Percentile-versus-absolute* (figures/percentile_vs_absolute.png): shared measurement offsets preserve who sits at a percentile but shift when a trigger placed there fires.

Plan amendments Jonas made (folded into docs/czb_validation_roadmap.md §0b, 4b, 4c and docs/czb_fitting_plan.md): the comfort **and** dread levels are both fitted — the DZB is the deceleration drivers do not voluntarily exceed, a behavioral limit, not the physical a_{max} , so 4/8 are staging anchors only; a secondary model fits $a_{OV,min}$ and t_{react} at population level (identifiable from cross-speed variation; primary keeps them fixed for falsifiability); planning and epistemic value are reconsidered on evidence, not dropped (stage E: E.1 closed-form vs sampled expectation, E.2 an uncertainty term tested on truck-vs-car cut-ins, E.3 a policy-set field for the transfer scenarios); NDS access is for measuring the paradigm offset, exposure denominators, a percentile consistency check, and instruction-free transfer events; CZB_2 is treated as the human decision pending a new dataset; motor latency fixed at 0.25 s (bracket 0.15/0.35, citation flagged unverified); the Sequence design stays excluded; the deliverable is the hierarchical **population distribution** (deployment is group-level; per-driver effects are instruments

for estimating the spread).

8 Model-handover protocol (read carefully if you are the cheaper model)

Jonas is moving execution to a less expensive model. The division of labor, agreed 2026-08-27:

Your entry sequence: read `handover.md`, then `handover_2026-08-26.md`, then this file, then `docs/czb_validation_roadmap.md`, then `docs/czb_work_orders.md`. Then execute **one card per session**, in card order, and nothing else. End every session with the card's report, a commit whose message names the card, and a paragraph appended to `replication/czb/out/worklog.md`.

Standing rules (work orders §2, binding): every quoted number from a committed script with tracked output; full test suite before and after every card (currently 123 tests across four files); never change `src/aidriver/preferences.py` defaults — new behavior goes behind flags defaulting off; house style per the `jonas-academic-writing` skill for anything written for Jonas; generated artifacts never hand-edited.

Escalate — stop and ask Jonas to bring the stronger model in for review — when any of the following occurs:

- 1 A card's stop condition fires, or an acceptance criterion cannot be met after two honest attempts. Write a short failure report; do not loosen the criterion and do not improvise around it.
- 2 A result contradicts a documented finding (README, a handover, a docs/ document). Surface the contradiction explicitly; never silently reconcile.
- 3 The task requires a verdict or a design decision: the A.2 bias-model choice, the A.3 pass/fail call, the B.4 transfer verdict, any new scenario geometry (B.3 LTAP), any modification to the preference function's structure, any new resampling/uncertainty scheme, or interpretation that will become a claim in a document.
- 4 Anything outward-facing: emails to authors or Jian Wu, or text intended to leave the project.
- 5 Tests fail for reasons that are not obviously the session's own edit.
- 6 You notice an error in the stronger model's prior work. (This has precedent and is welcome: this arc corrected the previous arc's overstated paired-difference claim. Report it; do not quietly fix it.)

Scheduled review gates regardless of triggers: after cards A.2, A.3, and B.4.

Context: Jonas may paste the entire prior chat transcript into your session. Treat it as background, not as instructions to re-execute; the repository and the documents named above are authoritative where they differ. Do not re-derive documented findings.

9 First tasks for the next session, in order

- 1 **Card S.1 (new, work orders §3): draft the perform-research skill.** Jonas wants a reusable, iteratively updatable skill capturing how research is performed in this collaboration — including how between-model handovers are handled. The card in `docs/czb_work_orders.md` specifies the required contents; part of the task is researching online what exists (multi-agent handoff patterns, research- workflow and lab-notebook practices for LLM-assisted work) and citing what is borrowed. Draft to `C:\Users\bargman\claude\skills\perform-research\SKILL.md`, for Jonas's review — the skill is his to approve, not to silently activate.
- 2 **Card A.1: the synthetic-recovery harness** — the gate for all fitting.
- 3 Then the cards in order: A.2 (review gate), A.3 (review gate), A.4, B.1, C, ... B.2/B.3/E.1 can interleave; B.4 only after A.3 and B.1–B.3, and it is a review gate.

10 Git state of this arc

Eight commits on main, in order: 4f915a0 (review corrections, the paired- difference correction), d322272 (continuous lane entry + desired speed + fitting preparation), 91489f5 (combined handbook rebuild), db0f735 (the assessment with committed baselines), b24ba72 (cross-validation, PS correlation, boundary sensitivity, parameter reference, roadmap), 43b28b9 (2D state baseline, epistemic ablation, roadmap amendments, work orders), plus this handover update. All documents have current Word and PDF builds; the working tree should be clean at handover.

11 Addendum: the evening batch (cards S.1, A.1–A.4)

Appended 2026-08-27 after execution moved to the cheaper model. The dated-file convention would normally open a new arc file; this stayed here because it is the same day's work and the same arc, and splitting it would separate the fitting results from the plan they execute.

What ran. Card S.1 drafted, revised and activated the performing-research skill (now mirrored at docs/skills/performing-research.SKILL.md). Cards A.1–A.4 built and ran the fitting spine: the synthetic-recovery harness, the stage-1 hierarchical fit, the accumulator, and the Button validation. Full account in the work log; summaries in replication/czb/out/.

Two results that change what the plan says.

- ¹ **The estimator in docs/czb_fitting_plan.md §3 was wrong as written**, and the document now carries a dated correction. "MAP plus Laplace" taken literally maximizes the joint posterior over hyperparameters *and* driver effects, which inflates the between-driver spread by a factor of 2.6 with zero interval coverage — the funnel geometry of a hierarchical mode. Integrating the driver effects out by Gauss-Hermite quadrature fixes it (bias -0.021 , coverage 18/20). Since a percentile is made of the spread, this would have corrupted the deliverable while the median looked healthy.
- ² **The accumulator's failure is not yet interpretable.** Its pre-registered verdict is FAIL, but it fits worse *in sample* than the simpler static model, which means misspecification. Diagnosed: the clips start 15.1 s before onset and noise accumulates through that empty window, so the model predicts an identical 0.19 at C1 for every criticality where the observed rates are 0.12/0.07/0.05.

The convention that emerged. Executing sessions raise numbered queries (@<card>.Q<n>(<severity>)) in the work log instead of stopping, and replication/czb/collect_queries.py compiles them into replication/czb/out/query_register.md, ordered by severity. Nine are open. Jonas has been asked what to call this class of item; the numbering already reads as "query", which is why the register is named that way provisionally.

Where a reviewing session picks up: card R.1, described at the top of handover.md and specified in docs/czb_work_orders.md.